DESIGN SYSTEMS

TECHNICAL DATA SHEET DRENACEM
PERMEABLE CONCRETE

DESCRIPTION

Permeable concrete is a cementitious conglomerate designed to create a system of interconnected porosity within the hardened mix, allowing water to pass through. This type of concrete is particularly effective for managing the flow of water on roads, letting the water run through the slab and thus avoiding dangerous pooling of water.

MIX

Permeable concrete is made up of cement, aggregates, water and admixtures. The main difference compared with conventional concrete lies in the proportioning of the aggregates. While traditional concrete uses several grading classes to obtain a closed structure, permeable concrete uses a single size, preferably single-sized (monogranular), to ensure an open structure that controls the hydraulic conductivity according to the maximum diameter of the aggregates. The use of uniformly sized aggregates also improves the appearance of the finished work.

Mixing the components produces an agglomerate of stone granules bound together by the cement paste. The absence of fine aggregate produces a highly porous concrete, characterised by interconnected voids that form channels ranging in size from 1 to 5 mm, thus determining the draining capacity of the structure.

AGGREGATES

Permeable concrete uses coarse aggregates with a variable maximum diameter, preferably single-sized, with sizes between 10 and 20 mm. Smaller aggregates create a less permeable structure, while aggregates with a maximum size of 20 mm or more promote greater permeability. However, a large diameter can reduce the contact points between the granules, compromising the mechanical strength. Natural, rounded aggregates are recommended for better bonding and drainage, avoiding elongated or flattened ones.

CEMENT

All types of cement that comply with the regulations in force can be used in the production of permeable concrete.

The binder content generally varies between a minimum of 280 kg/m^3 and a maximum of 430 kg/m^3, with typical values between 300 and 350 kg/m^3. The water/cement ratio (W/C) for permeable concrete is usually between 0,25 and 0,45. The water content can vary from a minimum of 100 l per cubic metre to a maximum of 130 l.

It is essential to dose the water correctly in order to avoid problems such as washing out of the cement paste or the formation of lumps. Excess water can lead to progressive clogging of the pores, while an insufficient quantity can make placing difficult.

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DRAINAGE VALUE

The main characteristic of permeable concrete is its hydraulic conductivity, i.e. its capacity to drain a given quantity of water from a surface. This property is closely related to the percentage of voids in the material, which usually varies between 15% and 30%. This percentage is known as effective porosity, defined as the ratio between the volume of water released by gravity from a completely saturated sample of draining material and the total volume of the sample.

Hydraulic conductivity is therefore influenced by the effective porosity, which depends on the size of the voids and on the system of channels connecting them. A higher void content increases the draining capacity of the concrete, but may reduce the mechanical strength because of the more open structure. Conversely, a lower void content reduces the draining capacity but improves the mechanical strength.

COMPRESSIVE STRENGTH

The quantity of water required for permeable concrete is crucial for obtaining the proper consistency during placing and finishing operations. However, this water demand is not decisive for the development of the mechanical strength of the material.

In permeable concrete, in addition to the water/cement ratio (W/C), there are other factors that influence the mechanical properties. Among these, the maximum diameter (Dmax) of the aggregates, the quantity of cement paste, the void content and the degree of interconnection between the voids themselves are key elements in achieving the desired mechanical strength.

The compressive strength of permeable concrete, measured according to the standard methods for conventional concrete, does not depend directly on the W/C ratio. It is instead influenced by the degree of compaction and by the cohesion between the granules. It is therefore advisable to reconsider the traditional methods laid down by the regulations in force for the preparation of specimens, in order to assess whether they are also applicable to permeable concrete.

ADMIXTURES

The performance of permeable concrete is strongly influenced by the mix design study, by the preparation of the mix, by the methods and times of laying, and by the environmental conditions. However, the physical-mechanical characteristics and durability of permeable concrete can be further improved through the use of appropriate chemical admixtures.

The performance of permeable concrete can be improved with chemical admixtures, such as: Polidre, a rubber-latex-based admixture that improves the bond between the aggregates, and Drena-25, a set retarder that extends the workability, while DS-F synthetic polypropylene fibres prevent surface wear and improve the bond.

Polidre: a liquid rubber-latex-based admixture specifically designed for the production of permeable concrete. This admixture improves the bond between the aggregate granules by increasing the binding capacity of the cement paste. It also improves the cohesion of the mix, facilitating placing operations and reducing the elastic modulus value.

The use of high dosages of Polidre ensures greater strength of the permeable concrete, making it more effective against freeze-thaw cycles, as well as controlling the action of de-icing salts.

Drena-25: the loss of moisture in the permeable concrete mix can result in a reduction of the workability time. Because of its open structure and low water content, permeable concrete tends to evaporate more quickly. To deal with this problem, a set retarder such as Drena-25 is used, which improves the working times.

MIX DESIGN

Several formulations are available

TABLE

Components	Mix 1	Mix 2	Mix 3
	kg/m^3	kg/m^3	kg/m^3
Cement	330	330	400
Gravel	1200 (grading 8/10)	1200 (grading 8/10)	1200 (grading 12/16)
Sand 0 - 3	-	150	100
Water	110	110	130
W/C ratio	0,33	0,33	0,32
Density	1635	1727	1850
Polidre admixture	3,5	3,5	4,5
Drena-25 admixture	2	2	2
DS-F	0,6	0,6	0,6
% Porosity	30,00%	23,00%	25,00%
Rcm	15	20	25
Drainage value	1200 mm/h	286 mm/h	727 mm/h

POURING

Permeable concrete, owing to the lack of fine materials in the mix and its poor workability, cannot be pumped; therefore, before proceeding with laying, it is important to make sure that the site can be accessed with suitable vehicles so that it can be unloaded properly.

Truck mixers used to transport permeable concrete should not carry a load of more than 6 cubic metres. The need not to fill the drum completely is due to the fact that the concrete is drier and may require a longer unloading time, but also because mixing a large mass of permeable concrete places excessive strain on the mixer's engine.

A permeable concrete mix should be completely unloaded within one hour of the start of mixing.

For a successful permeable pavement, it is essential that the substrate is free of irregularities, misalignments and depressions, so as not to affect the flatness of the pavement.

Permeable concrete must be compacted by rolling, preferably with vibratory screed pavers and "roller screeds". The compaction load must come only from the weight of the roller, since excessive pressure could cause the voids to collapse and reduce the draining capacity. However, insufficient compaction could make the pavement less durable and prone to delamination.

It is advisable to complete the laying operations within 15 minutes of unloading the concrete, in order to avoid as far as possible compaction problems related to rapid hardening and surface evaporation of the water.

JOINTS

Permeable concrete has, on average, a lower drying shrinkage than traditional concrete, both because of its low cement content and because of its low water content; however, contraction joints should be provided to control the cracks that could form on the surface. The joint spacing can be established by applying the general rule L = (18 x h)+100, where L is the length of the slab and h the thickness in cm.

The joints should be made perpendicular to the direction of laying, immediately after the roller has passed but before the curing operations.

CURING

As with conventional concrete, curing of permeable concrete is essential for the correct hydration of the cement binder. The open structure that characterises permeable concrete exposes it to greater evaporation, since the specific surface area exposed is greater. For this reason, moist curing must begin as soon as possible. Proper curing begins before the concrete is placed, by saturating the substrate so that it does not draw moisture from the concrete. Surface curing should start no later than 20 minutes after placing, unless specific agreements have been made between the parties. The surface of the concrete must be wetted or moistened with water and immediately covered with polyethylene sheets for a period of at least 7 days.

INSTALLATION

Permeable concrete can be laid on an existing sub-base layer if this consists of loose material such as sand and gravel. In this case the sub-base should first be compacted so as to create a stable and uniform surface. In the case of a clayey or silty sub-base (low permeability), the most suitable solution is to scarify the surface to improve infiltration, lay a layer of non-woven geotextile and then a layer of loose material, the thickness of which will vary according to the hydraulic capacity required of the pavement.

Optional drainage systems may be required by the designer if the hydrological conditions of the site are such that they do not allow the pavement to discharge water adequately. Where the sub-base has low permeability, the surface of the sub-base can be covered with a waterproof membrane and a system of channels can be laid within the layer of loose material to collect the water and convey it outside the pavement.

Once the water conveyance system has been installed and the draining sub-base layer has been completed, the permeable concrete can be laid.

The layer of loose aggregate must always be thoroughly wetted before the concrete is placed, to prevent the dry material from drawing water from the fresh concrete and to protect it from excessively rapid setting and drying.

The thickness of permeable concrete, when used for pavements, is generally between 15 - 20 cm. The lower thicknesses are used for pedestrian areas and increase progressively in relation to traffic. The use of permeable concrete for footpaths or cycle paths, on the other hand, does not require a minimum thickness.

PIGMENTATION

Drenacem can be coloured with colour pigments (DS-Ossido) to be added to the concrete mass. The proportion of DS-Ossido ranges from 3% - 5% of the weight of the cement.

PROTECTION OF THE DRENACEM FLOOR

The permeable floor can be protected with DS-CS resin, applied by spray or roller, to improve its durability over time. Average consumption 250 – 300 g/m^2.

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof according to Design Systems methods. The procedures applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. For uses other than those indicated, it is nevertheless advisable to carry out preliminary tests. In any case, Design Systems Srl accepts no liability for damage or defects caused by the use of its products, as the conditions of use are beyond its control. It also informs that its technical service is available to customers for any information concerning the correct use of its products.